

Lecture 10: Diagonalization and Undecidability

203 Review: *Countability*

- A (total) function $f: A \rightarrow B$ maps each element $a \in A$ to an element $f(a) \in B$.
- A function is 1-to-1 (aka *injective*) if
- If a 1-to-1 function $f: A \rightarrow B$ exists, then $\text{_____} \leq \text{_____}$.
- A and B have the *same* cardinality, i.e., $|A| = |B|$, if we have both _____ and _____ .
- A set is **countable** if it is not larger than the naturals $\mathbb{N} = \{0, 1, 2, \dots\}$
- In other words, it is possible to **list every element** of the set.
- Bonus: If an onto function $f: A \rightarrow B$ exists, then $\text{_____} \leq \text{_____}$.

Is ____ a countable set? Why?

The integers from 1 to 10?

The non-negative even numbers?

The integers? (positive, 0, and negative)

The rationals?

The reals in $(0,1)$?

Diagonalization

Theorem: the set of reals in $(0,1)$ is _____.

Suppose for contradiction we could write a list of all the reals in $(0,1)$.

Any $r \in (0,1)$ can be represented as _____

$r_i = i^{\text{th}}$ real number in your list.

r_1	0.	5	6	1	2	8	...		...
r_2	0.	4	4	4	4	4	...		...
r_3	0.	1	2	3	4	5	...		...
r_4	0.	8	7	2	1	6	...		...
...							...		...
							...		...

Define the real number r^* to be:

$$r^*(i) =$$

Then r^* must appear in the list somewhere! Suppose $r^* = r_j$

What is _____?

Our goal is to find a real in $(0,1)$ that's not in this supposed list!



There are _____ many Languages!

- Suppose (for contradiction) _____
We could list them.
- Construct a language L' , where _____
- Because (by assumption) every language on $\{0,1\}$ is in the list, L' must be some L_j
- Is _____? Contradiction!

[illegible]

Strings, Languages, TMs (un)Countable?

1. How many strings are there on the alphabet $\{0,1\}$? $|\{0,1\}^*| =$

2. How many languages are there on the alphabet $\{0,1\}$?

3. How many Turing Machines are there?

What do the answers to 2 and 3 tell us?

Who's cutting whose hair around here?

- There's a lot of cutting hair going on in this town!
- How can we keep track of who's cutting whose hair?
 - *Maybe with, like, a "hairstresser-hairstressee" matrix?*

		Hairstresses						
Hairstresses		A	B	C	D	E		
	A	☒	✗	✗	✗	✗		
	B	✗	✗	☒	✗	☒		
	C	✗	☒	✗	✗	✗		
	D	☒	✗	✗	☒	✗		
	E	✗	✗	☒	✗	✗		
	Barbara							

Translate the Barber Paradox into a story about TMs

- Your translation guide:

- “**people, Barbara**” → _____
- “**cut hair**” → _____
- “**hairedressee**” → _____

“**Barbara cuts the hair** of those people—and only those people—
who do not **cut the hair** of themselves”

“ _____ of those _____—and only those _____—
who do not _____ of _____ ”

- Why, precisely, is the existence of the _____ a problem?
Is there still a paradox?